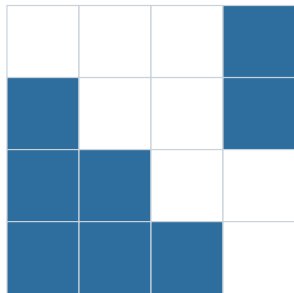


## A LEAF STATISTICS

Representative slice of a  $4 \times 4 \times 4$  occupancy mask

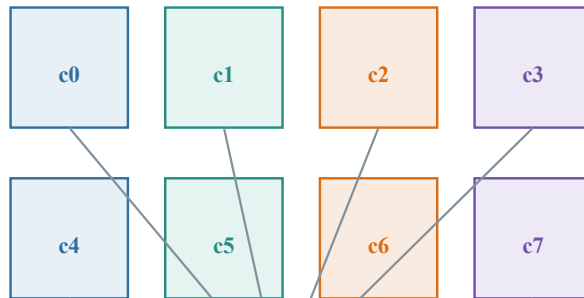


Shift the mask along  $\pm X$ ,  $\pm Y$  and  $\pm Z$ ,  
then use bit operations and population counts.

$A_d$ : exposed surface area  
 $B_d$ : occupied boundary-slab area

## B BOTTOM-UP AGGREGATION

Eight child descriptors



Sum child areas and retain only children  
touching each parent boundary.



Subtract sibling interface overlap  
 $I_{ij} \approx B_i B_j / S_c^2$

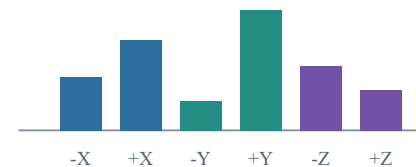
## C NODE DESCRIPTOR

Remove the node hull already represented by the box  
normal:

$$r_d = \text{clamp}((A_d - B_d) / S^2, 0, 1)$$



Six-direction internal-relief histogram



$$C_a = (A_{-a} + A_{+a}) / (2S^2)$$

$$P_a = [C_a < T_{\text{cov}}]$$



Memoised by unique node index  
One 32-bit result is written inline.